The Tandem Van de Graaff Accelerator

Step-by-Step with Diagrams

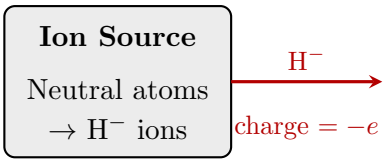
Supplementary to Mann, Chapters 7–8

Key Idea

A **single-stage** Van de Graaff accelerates a charged particle through a potential V once, giving it kinetic energy $K = qV$. A **tandem** Van de Graaff uses a *charge-stripping trick* so the *same* terminal voltage V accelerates the particle *twice*, yielding $K = (1 + n)\,eV$ for an ion stripped to charge $+ne$. This doubles (or more) the energy without increasing the voltage—which is the engineering bottleneck.

Step 1: Produce Negative Ions

The process begins with a negative-ion source. Neutral atoms (e.g. hydrogen) are given an *extra* electron, producing H^- ions (a proton with two electrons).



Why negative?

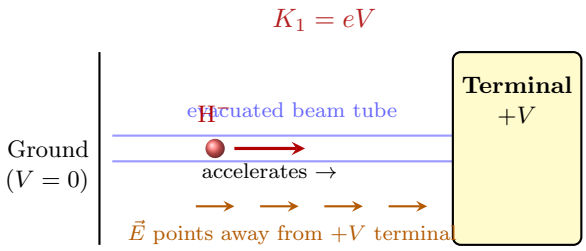
Starting with a *negative* ion is the entire trick. A negative ion is *attracted toward* a positive high-voltage terminal—so it accelerates *into* the terminal. After stripping, the now-positive ion is *repelled away* from the same terminal—so it accelerates again on the way *out*.

Step 2: First Acceleration Stage (Low-Energy End → Terminal)

The H^- ion enters a long evacuated tube. At the center of the machine sits a **high-voltage terminal** held at potential $+V$ (typically 10–25 MV) by a Van de Graaff charging belt or chain.

Since the ion has charge $q = -e$ and moves toward higher potential, the electric force accelerates it. It gains kinetic energy:

$$K_1 = eV$$

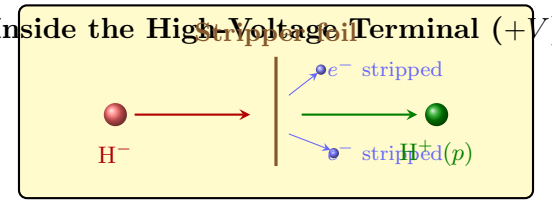


Step 3: Charge Stripping at the Terminal

Inside the terminal, the beam passes through a **stripper**—either a thin foil (typically carbon, a few μg/cm^2) or a gas cell. This knocks off *all* the electrons. For hydrogen:

$$\mathrm{H}^- \longrightarrow \mathrm{H}^+ + 2e^- \quad \text{(proton, charge } +e\text{)}$$

For heavier ions (e.g. carbon, oxygen), multiple electrons can be stripped, yielding charge states $+ne$ where n can be large.



The sign flip is everything

Before the stripper, the particle had charge $-e$ and was *attracted toward* $+V$. After the stripper, it has charge $+ne$ and is *repelled away from* $+V$. The same terminal voltage now accelerates it a *second time* in the opposite direction—out of the machine.

Step 4: Second Acceleration Stage (Terminal → Ground)

The now-positive ion ($+ne$) travels from the terminal at $+V$ back down to ground potential $V = 0$. It gains additional kinetic energy:

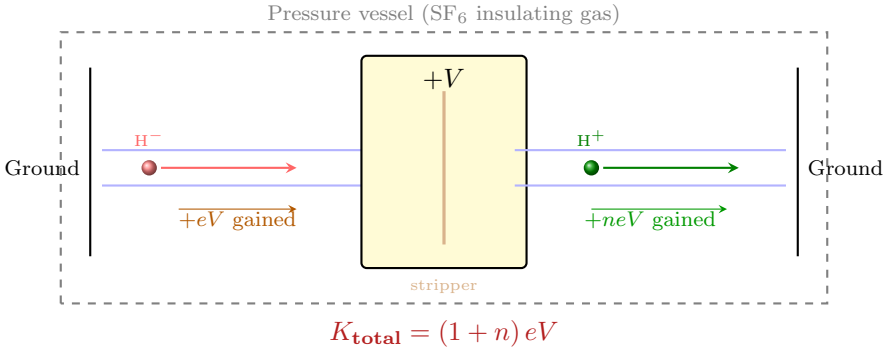
$$K_2 = neV$$

The **total kinetic energy** is:

Total Energy from Tandem Van de Graaff

$$\boxed{K_{\text{total}} = K_1 + K_2 = eV + neV = (1+n)\,eV}$$

For protons ($n = 1$): $K = 2eV$. For fully stripped $^{12}\mathrm{C}^{6+}$: $K = 7eV$.



Step 5: The Complete Picture

Putting it all together, the full sequence is:

1. **Negative-ion source** produces H^- (or other negative ions) at ground potential.
2. **First acceleration**: H^- accelerated from ground toward $+V$ terminal; gains eV.
3. **Stripping**: electrons removed at terminal; ion becomes $+ne$.
4. **Second acceleration**: $+ne$ ion repelled from $+V$ terminal back to ground; gains neV.
5. **Beam transport**: the high-energy beam is steered by analyzing magnets to the target.

Why "Tandem"?

The name comes from the two acceleration stages arranged in *tandem* (one after the other), using the same terminal voltage. A single Van de Graaff limited to 10 MV gives 10 MeV protons; a tandem gives 20 MeV protons from the same voltage.

Engineering note: insulating gas

The entire accelerator column sits inside a pressure vessel filled with SF_6 (sulfur hexafluoride) at several atmospheres. SF_6 is an excellent electrical insulator, suppressing sparking and allowing terminal voltages up to ~25 MV.

Comparison: Single vs. Tandem

	Single-stage VdG	**Tandem VdG**
Source ion	positive (H^+)	negative (H^-)
Accelerations	1	2
Proton energy	eV	$2eV$
Heavy ion ($+ne$) energy	neV	$(1+n)\,eV$
Terminal at center?	No (at top/end)	Yes
Needs stripper?	No	Yes